Project Design Phase
Proposed Solution Template

Date	28 June 2026
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Proposed Solution Template:

Project team shall fill the following information in the proposed solution template.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Farmers in India lack access to a data-driven tool for selecting the right crop based on their specific soil and environmental conditions. Without scientific guidance on Nitrogen (N), Phosphorous (P), Potassium (K), temperature, humidity, pH, and rainfall, farmers resort to guesswork, leading to poor yields, resource wastage, and financial losses. Agricultural researchers and policymakers also lack a unified platform to analyze crop-environment patterns for evidence-based decision making.
2.	Idea / Solution description	OptiCrop is an ML-powered web application built using Python (Flask, Scikit-learn) that accepts key soil and environmental parameters as input and recommends the most suitable crop for maximum yield. Key features include: (1) Smart Crop Recommendation Engine – predicts optimal crop from N, P, K, temperature, humidity, pH and rainfall inputs; (2) Crop Suitability Assessment – evaluates whether conditions are suitable for a specific crop; (3) Data Visualization Dashboard – Matplotlib and Seaborn-based charts for researchers and policymakers; (4) User-friendly Flask web interface accessible without technical expertise.
3.	Novelty / Uniqueness	OptiCrop uniquely integrates 7 critical agri-environmental parameters (N, P, K, temperature, humidity, pH, rainfall) into a single ML-based recommendation engine – a combination not available in free, accessible tools for Indian farmers. Unlike generic crop guides, OptiCrop provides farm-specific, real-time predictions. It serves three distinct user groups (farmers, researchers, policymakers) from one platform, and uses Scikit-learn classification models trained on real agricultural datasets to ensure high accuracy.
4.	Social Impact / Customer Satisfaction	OptiCrop directly benefits small and medium-scale farmers by reducing crop failure risk and improving resource efficiency (less fertilizer waste, better water use). For India's ~100 million farming households, even a modest yield improvement translates into significant income gains and food security. Researchers gain a data analysis tool for crop-environment studies, while policymakers can design targeted, evidence-based agricultural schemes. The platform promotes sustainable farming and contributes to UN SDG 2 (Zero Hunger) and SDG 12 (Responsible

		Production).
5.	Business Model (Revenue Model)	Freemium SaaS Model: Basic crop recommendation is free for farmers via the web app, driving mass adoption. Premium tier (subscription) for agri-businesses, cooperatives, and research institutions offers bulk analysis, API access, and detailed reports. B2G (Business to Government): License the platform to state agriculture departments for integration into farmer advisory schemes (e.g., PM-KISAN). Data Monetization: Aggregated, anonymized crop-environment data sold as insights to agri-input companies and seed manufacturers.
6.	Scalability of the Solution	OptiCrop is built on Python and Flask, which are cloud-deployable on AWS, Azure, or GCP with minimal infrastructure cost. The ML model can be retrained with expanded regional datasets to cover more crops and geographies across South Asia and globally. The modular architecture allows adding new input parameters (e.g., satellite imagery, IoT soil sensors) and new output modules (fertilizer dosage, irrigation schedule). Mobile app development (Android/iOS) will further expand reach to rural farmers with smartphones.